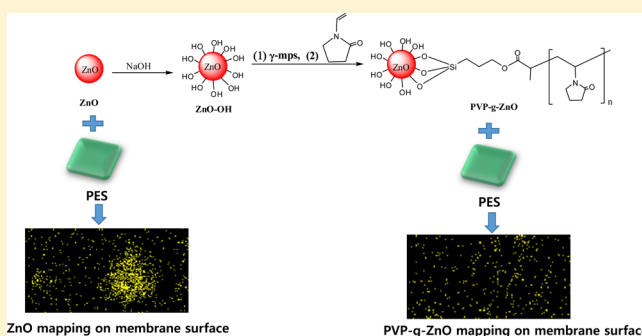


Antibacterial and Hydrophilic Characteristics of Poly(ether sulfone) Composite Membranes Containing Zinc Oxide Nanoparticles Grafted with Hydrophilic Polymers

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ABSTRACT: An ultrafiltration membrane with antibacterial effectiveness and high water flux was prepared from poly(ether sulfone) (PES) and zinc oxide (ZnO) nanoparticles grafted with poly(1-vinylpyrrolidone) (PVP-g-ZnO) and poly(1-vinylpyrrolidone-co-acrylonitrile) (P(VP-AN)-g-ZnO). The PES membrane did not exhibit antibacterial activity, whereas the PES membranes containing more than 0.5 wt % of ZnO nanoparticles exhibited antibacterial activity, regardless of the polymer grafted onto the ZnO. The solute rejection measured with the PES/P(VP-AN)-g-ZnO membrane was nearly identical to that measured with the PES membrane. The water flux of the PES membrane involving polymer-grafted ZnO nanoparticles was higher than that of the PES membrane containing pristine ZnO nanoparticles when the ZnO concentration in the membrane was fixed. The PES/PVP-g-ZnO and the PES/P(VP-AN)-g-ZnO membranes exhibited nearly identical water flux. Membranes exhibiting antibacterial effectiveness, improved water flux, and antifouling characteristic without changes in solute rejection could be produced by incorporating P(VP-AN)-g-ZnO.



1. INTRODUCTION

Many water treatment technologies are being developed to supply high quality water and reclaim wastewater in growing chemical and manufacturing industries.^{1–4} Among these technologies, membrane processes have been increasingly applied to water treatment owing to the reduced cost, high efficiency, and stable permeate quality. However, application of membrane technology to water treatment is limited because of membrane fouling. A decline in the water flux of the membrane occurs with increasing operating time owing to the foulant accumulation on the membrane.

Membranes produced from poly(ether sulfone) (PES) for ultrafiltration processes show excellent hydrolytic and thermal resistance as well as outstanding mechanical strength.^{5,6} However, because of the hydrophobic nature of the PES, the PES membrane shows a low water permeation compared to other hydrophilic membranes, and its water flux declines continuously with operating time owing to the foulant accumulation on the membrane surface and in pores.^{7–11} Furthermore, biofouling due to the attachment of the microorganisms and their growth on the membrane also brings about physical obstruction to flow.^{2,12–22} Since foulants attached to the membrane deteriorate the membrane performance and are difficult to remove, membranes having antifouling characteristics should be developed by incorporating an intrinsic hydrophilic nature and antibacterial properties into the membranes.

Considerable effort has been expended to enhance the performance of PES membranes. Modifications of PES

membranes via chemical treatment, UV-grafting, coating, and plasma treatment are the approaches that have been used to provide a hydrophilic nature.^{23–26} PES membranes containing hydrophilic polymers and inorganic and metallic nanoparticles including titanium oxide, silica, aluminum oxide, zinc oxide, silver, copper, carbon nanotubes (CNT), and graphene oxide (GO) have been developed.^{2,18,19,27–42} Basri et al. prepared PES composite membranes containing silver nanoparticles and PVP to enhance the hydrophilicity and antibacterial characteristics of the membranes.³⁵ The hydrophilic nature of these membranes gradually decreased due to the leaching of the PVP component during the filtration process. To retain the water-soluble polymer in the PES matrix, silica nanoparticles grafted with a water-soluble polymer were incorporated into the PES matrix.^{33,34} However, these membranes did not show antibacterial effectiveness. PES membranes combined with polyethylenimine grafted GO exhibited improved antibacterial activity with a slight loss of water flux.³³ Sawai et al. evaluated the inhibition effect of 26 ceramic powder slurries on the growth of bacteria.⁴⁴ The growth of the test bacteria was inhibited by 10 ceramic powders such as MgO, CaO, and ZnO. There are also other reports documenting the antibacterial activity of ZnO nanoparticles.^{45,46} It is known that the antibacterial characteristic of ZnO originates from the hydrogen

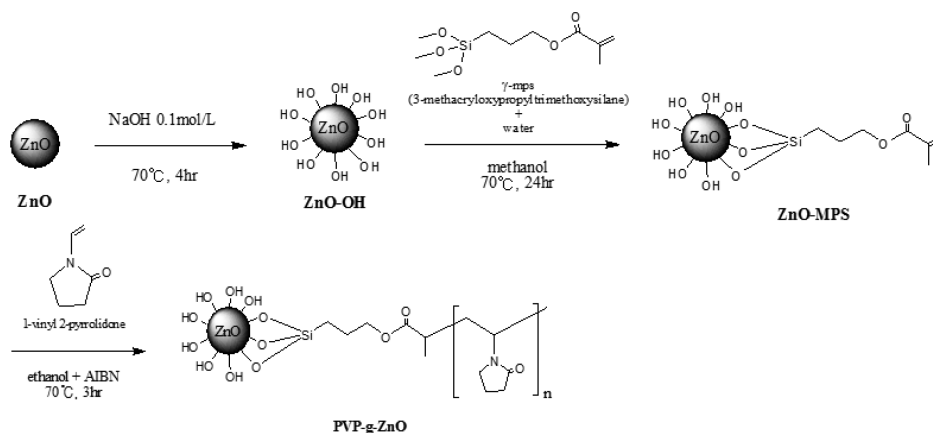
Received: April 19, 2016

Revised: June 28, 2016

Accepted: July 1, 2016

Published: July 1, 2016

Scheme 1. Synthetic Route of PVP-Grafted ZnO Nanoparticles



peroxide generated at the ZnO surface.⁴⁶ The PES membranes employing ZnO nanoparticles coated with PVP and those assembled with ZnO nanoparticles on the membrane were also studied to enhance water permeability and antifouling performance.^{37,38} Research is still ongoing to develop modified PES membranes exhibiting improved long-term performance.

In this study, PES membranes are modified by incorporating zinc oxide (ZnO) nanoparticles grafted with hydrophilic polymers. Herein, ZnO nanoparticles grafted with hydrophilic polymers are incorporated into the PES matrix with the expectation that this combination can simultaneously impart antibacterial and hydrophilic properties to the PES membrane. The performance, antibacterial characteristic, and fouling resistance of the PES membranes containing ZnO nanoparticles were also compared to those of the unmodified PES membrane.

2. MATERIALS AND PROCEDURES

2.1. Materials. Poly(ether sulfone) (PES, grade: Veradel 3000P, $M_w = 57\,000$ g/mol) supplied from Solvay Korea and zinc oxide (ZnO) nanoparticles (average diameter: 50 nm) purchased from Sigma-Aldrich were used. 3-Methacryloxypropyltrimethoxysilane (γ -MPS), utilized as a reactant with the hydroxyl-terminated ZnO, 1-vinylpyrrolidone (VP), and acrylonitrile (AN) monomers were procured from Sigma-Aldrich. 2,2'-Azobis(2-methylpropionitrile) (AIBN), employed as an initiator for the radical polymerization, was purchased from Junsei Chemical Co. (Japan). Isopropyl alcohol, 1-methyl-2-pyrrolidone (NMP), and sodium hydroxide (NaOH) were also supplied by Sigma-Aldrich. Polyoxyethylene glycol alkylether ($H(CH_2)_{16}O(CH_2CH_2O)_8H$, $C_{16}E_8$) used as a foulant was provided by Nikko Chemicals (Japan).

2.2. Synthetic Route Polymer-Grafted ZnO Nanoparticles and Their Characterization. As shown in Scheme 1, polymer-grafted ZnO nanoparticles were synthesized by reacting vinyl monomers with ZnO nanoparticles functionalized with γ -MPS.³³ Pristine ZnO nanoparticles (5 g) were first added to an aqueous solution (200 mL) of NaOH (0.5 mol/L) and then mixed for 4 h at 70 °C to produce ZnO nanoparticles functionalized with hydroxyl groups (ZnO-OH). To synthesize ZnO nanoparticles coupled with γ -MPS (ZnO-MPS), ZnO-OH (5 g) was dispersed in a solution containing deionized water (400 mL), methanol (100 mL), ammonium hydroxide (0.5 mol), and γ -MPS (5 g) and then reacted at 70 °C for 24 h. After separating ZnO-MPS from unreacted γ -MPS by using an ultracentrifuge, the ZnO-MPS was further

purified by washing with isopropyl alcohol (3×180 mL). The resulting product was dried at 70 °C in a vacuum oven for 24 h to remove residuals.

Hydrophilic polymer-grafted ZnO nanoparticles were produced by a radical polymerization. To synthesize PVP-grafted ZnO nanoparticles (PVP-g-ZnO), ZnO-MPS (5 g) was dispersed in an ethanol (200 mL) solution containing VP (100 mL) and AIBN (0.5 wt % of VP); the reaction was performed at 70 °C for 3 h under reflux, and then terminated by cooling in a water bath. The resulting product, recovered by ultracentrifugation, was washed with isopropyl alcohol ($300\text{ mL} \times 3$) to separate PVP-g-ZnO from the residual reactants and polymer ungrafted onto the ZnO nanoparticles. The product was ultimately dried at 70 °C in a vacuum oven for 24 h. To produce the P(VP-AN) grafted ZnO nanoparticles (P(VP-AN)-g-ZnO), ZnO-MPS (5 g) dispersed in ethanol (250 mL) was reacted with a VP and AN mixture (150 g, VP/AN ratio = 98/2) using AIBN (0.5 wt % of monomers) as an initiator at 70 °C for 1 h with refluxing. Monomer conversion was controlled at approximately 10 wt % to produce the copolymer having the desired composition.^{29,47}

The structure of the polymer-grafted ZnO nanoparticles was characterized by using Fourier transform infrared (FT-IR, Magna 750, Nicolet, USA) analysis. Thermogravimetric analysis (TGA, model, TGA-2050, TA Instruments, USA) was performed to estimate the amount of polymer grafted onto the ZnO nanoparticles. TGA analyses were performed at a heating rate of 10 °C/min at a nitrogen condition. The carbon, nitrogen, and oxygen contents of the P(VP-AN) were analyzed by using an element analyzer (EA, Flash 2000, CE Instrument, Italy) to estimate the AN content in the P(VP-AN) polymer.

2.3. Membrane Fabrication. Asymmetric membranes for ultrafiltration were prepared by a nonsolvent induced phase separation (NIPS) process.^{5,28} A mixture of PES and NMP was heated at 80 °C for 10 h until PES was completely dissolved in NMP, and pristine ZnO or the polymer-grafted nanoparticles were dispersed in the PES solution under sonication at 30 °C for 1 h. Degassing was performed in a vacuum oven at 30 °C for 2 h. The PES solution was poured on a nonwoven polyester fabric and then cast with a doctor blade having a thickness of 100 μm . The cast solution was subsequently precipitated in a water bath for 24 h. The compositions of the casting solutions and the membranes and their abbreviations are presented in Table 1. The numbers in the abbreviations indicated ZnO wt % in the membrane.

Table 1. Composition of the Membranes and Their Abbreviation

solution composition (wt %)	membrane composition (wt %)	abbreviations of membranes
NMP/PES = 80/20	PES	PES
NMP/PES/pristine ZnO = 79.8/20/0.2	PES/pristine ZnO = 99/1	PES/pristine ZnO-1
NMP/PES/pristine ZnO = 79.4/20/0.6	PES/pristine ZnO = 97/3	PES/pristine ZnO-3
NMP/PES/pristine ZnO = 78.9/20/1.1	PES/pristine ZnO = 95/5	PES/pristine ZnO-5
NMP/PES/pristine ZnO = 78.5/20/1.5	PES/pristine ZnO = 93/7	PES/pristine ZnO-7
NMP/PES/pristine ZnO = 77.8/20/2.2	PES/pristine ZnO = 90/10	PES/pristine ZnO-10
NMP/PES/PVP-g-ZnO = 79.8/20/0.2	PES/PVP-g-ZnO = 99/1	PES/PVP-g-ZnO-1
NMP/PES/PVP-g-ZnO = 79.4/20/0.6	PES/PVP-g-ZnO = 97/3	PES/PVP-g-ZnO-3
NMP/PES/PVP-g-ZnO = 78.9/20/1.1	PES/PVP-g-ZnO = 95/5	PES/PVP-g-ZnO-5
NMP/PES/PVP-g-ZnO = 78.5/20/1.5	PES/PVP-g-ZnO = 93/7	PES/PVP-g-ZnO-7
NMP/PES/PVP-g-ZnO = 77.8/20/2.2	PES/PVP-g-ZnO = 90/10	PES/PVP-g-ZnO-10
NMP/PES/P(VP-AN)-g-ZnO = 79.8/20/0.2	PES/P(VP-AN)-g-ZnO = 99/1	PES/P(VP-AN)-g-ZnO-1
NMP/PES/P(VP-AN)-g-ZnO = 79.4/20/0.6	PES/P(VP-AN)-g-ZnO = 97/3	PES/P(VP-AN)-g-ZnO-3
NMP/PES/P(VP-AN)-g-ZnO = 78.9/20/1.1	PES/P(VP-AN)-g-ZnO = 95/5	PES/P(VP-AN)-g-ZnO-5
NMP/PES/P(VP-AN)-g-ZnO = 78.5/20/1.5	PES/P(VP-AN)-g-ZnO = 93/7	PES/P(VP-AN)-g-ZnO-7
NMP/PES/P(VP-AN)-g-ZnO = 77.8/20/2.2	PES/P(VP-AN)-g-ZnO = 90/10	PES/P(VP-AN)-g-ZnO-10

2.4. Membrane Characterization. The performance of the ultrafiltration membranes was estimated by using a flat-sheet cross-flow ultrafiltration test cell (active membrane area of 19.63 cm²; internal diameter, 50 mm; cell volume, 196 mL) as described previously.³³ The permeate flux and the solute rejection were evaluated at 30 °C, 3 × 10⁵ Pa, and a flow rate of 0.7 L/min. An aqueous solution containing 1000 ppm of polyethylene glycol (PEG, weight-average molecular weight, 40 000 g/mol; molecular weight distribution = 1.03) was used as the feed solution. Both the retentate and permeate solution were recycled to the feed tank to kept the concentration of the feed solution. The membranes were first pressurized with the PEG solution at 3 × 10⁵ Pa for 30 min, and then the permeate flux was measured. The PEG content in the permeate was estimated using a refractometer (model RI-2031, Jasco, Japan). Five membranes were tested and the average was recorded.

To evaluate the membrane hydrophilicity, the contact angle between a water droplet and the membranes was measured with a contact angle analyzer (Rame-Hart, model:100-00-(115/220)-S). The contact angles reported represent the average of 10 measurements for each membrane. The surface and cross-sectional morphologies of the membranes were observed with a field emission scanning electron microscope (FE-SEM, model: Sigma, Carl Zeiss, Germany). Leaching of ZnO nanoparticles from the membranes was evaluated with an inductively coupled plasma spectrometer (ICP, Jobin Yvon, model JY-Ultima-2, France). The PES/ZnO membranes were immersed into a water bath and sonicated at 0.3 KW (50 Hz) for 2 h. The ZnO concentration in the water was determined with ICP spectroscopy (wavelength, 213.856 nm) after sonication.

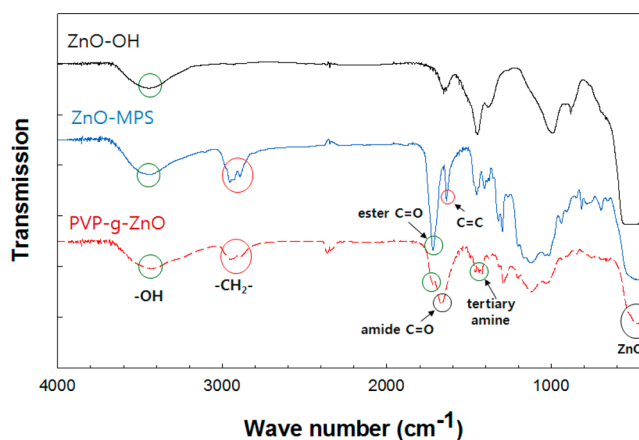
The antibacterial effectiveness of the membranes was evaluated according to the JIS Z 2801.⁴⁸ *Escherichia coli* (*E. coli*, ATCC 8739) and *Staphylococcus aureus* (*S. aureus*, ATCC 6538P) were used as the Gram-negative and Gram-positive bacteria, respectively. The membrane was cut into a square of 50 mm and placed in a sterilized Petri dish. The test inoculum was instilled onto the membrane surface and covered with a square film of 40 mm. The test inoculum was incubated at a temperature of 35 °C and a relative humidity of 90% for 24 h. The antibacterial activity was calculated according to eq 1.

$$\text{antibacterial activity} = \log \left(\frac{M_b}{M_c} \right) \quad (1)$$

where M_b and M_c are the average number of bacteria (colony forming units, CFU) on the control test samples and membrane samples after 24 h of incubation, respectively. An aqueous solution containing C₁₆E₈ was used to explore the flux decline with the operating time owing to membrane fouling. Note that feed solution contains 100 times the critical micelle concentration (CMC) of C₁₆E₈ (CMC of C₁₆E₈ = 1 × 10⁻⁶ mol/L).

3. RESULTS AND DISCUSSION

3.1. Characterization of Polymer-Grafted ZnO Nanoparticles. Figure 1 shows the FT-IR spectra of ZnO–OH,

**Figure 1.** FT-IR spectra of ZnO–OH, ZnO–MPS, and PVP-g-ZnO.

ZnO–MPS, and PVP-g-ZnO. A peak originated from ZnO was observed at 442 cm⁻¹ regardless of the surface treatment of the ZnO nanoparticles. Peaks originated from γ-MPS, methylene (2850–3000 cm⁻¹), carbonyl groups (1710 cm⁻¹), and carbon–carbon double bonds (1610 cm⁻¹) were observed in the ZnO–MPS spectrum. The stretching peaks associated with PVP, tertiary amine groups (1462 and 1424 cm⁻¹), and amide groups (1670 cm⁻¹), were observed in the PVP-g-ZnO spectrum. Notably, no peaks corresponding to the C=C bonds were revealed in the PVP-g-ZnO spectrum. Such a finding indicates that PVP-g-ZnO was fabricated by the coupling reaction between γ-MPS and the VP monomer and the radical polymerization among the VP monomers.

Figure 2 shows the FT-IR spectra of VP, AN, and P(VP-AN)-g-ZnO. The peak originating from the nitrile groups (2243 cm⁻¹) and peaks originating from PVP were observed in the P(VP-AN)-g-ZnO spectrum. These results demonstrate that P(VP-AN)-g-ZnO was also formed. The AN content of the copolymer was determined using an element analyzer. The AN

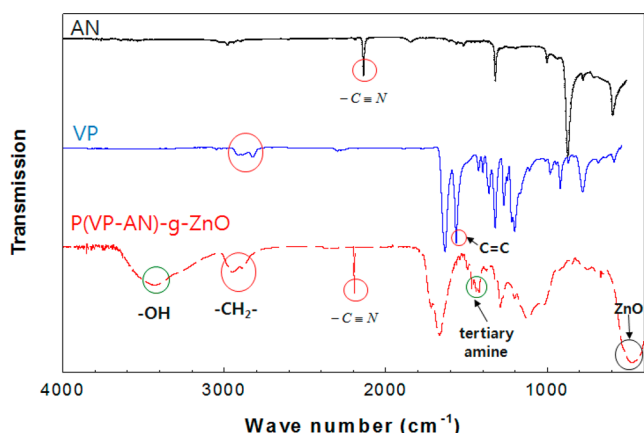


Figure 2. FT-IR spectra of AN, VP, and P(VP-AN)-g-ZnO.

content of P(VP-AN) grafted onto the ZnO, which was estimated by assuming that AN content of P(VP-AN) formed in the ethanol phase is equal to that of P(VP-AN) grafted onto the ZnO, was 10 wt %. It is known that PES forms miscible blends with P(VP-AN) copolymers containing AN from 8 wt % to 36 wt %, regardless of the blend composition.³⁵

TGA was used to measure the amount of polymer grafted to the ZnO nanoparticles. Figure 3 shows the TGA thermograms

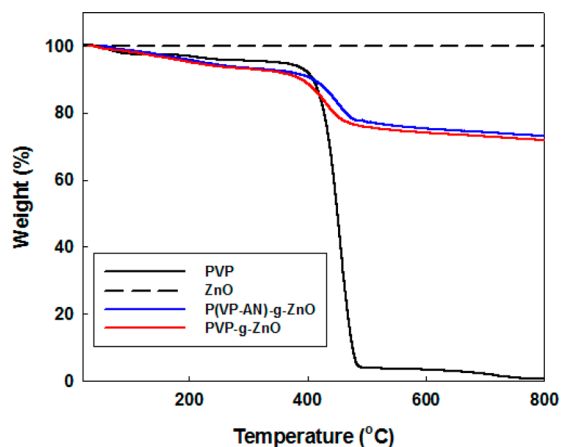


Figure 3. Thermal degradation behavior of pristine ZnO, PVP, P(VP-AN)-g-ZnO, and PVP-g-ZnO observed with TGA.

obtained for pristine ZnO, PVP, and PVP-g-ZnO. PVP grafted to ZnO underwent thermal degradation in the temperature range of 390–480 °C. Furthermore, P(VP-AN)-g-ZnO exhibited thermal degradation behavior similar to that of PVP-g-ZnO. The losses in mass observed for PVP-g-ZnO and P(VP-AN)-g-ZnO due to thermal degradation of the polymer were 16.5 and 15.2 wt %, respectively.

3.2. Changes in Membrane Performance. The water flux and PEG rejection of the membranes were examined as shown in Figure 4. The water flux of the membrane increased with increasing ZnO content regardless of the surface treatment of the ZnO nanoparticles. Errors in the water flux did not exceed $\pm 5\%$. At a fixed ZnO content, the water flux of the PES/PVP-g-ZnO (or PES/P(VP-AN)-g-ZnO) membrane was greater than that of the PES/pristine ZnO membrane. The water flux of PES/PVP-g-ZnO was nearly identical to that of the PES/P(VP-AN)-g-ZnO when the filler content was less than or equal to 5 wt %, but the water flux of the former was slightly higher

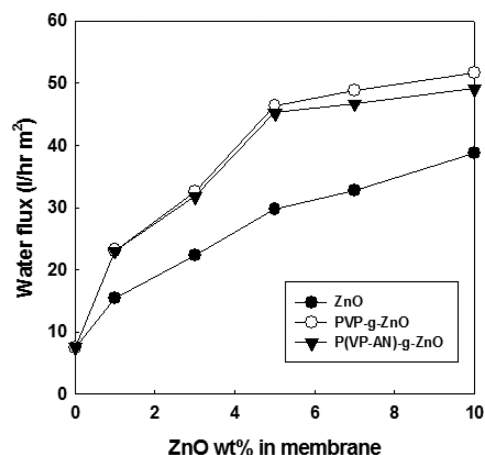


Figure 4. Changes in the water flux of the PES/pristine ZnO, PES/PVP-g-ZnO, and PES/P(VP-AN)-g-ZnO membranes with the ZnO content in the membrane.

than that of the latter when the filler content was higher than 5 wt %. Figure 5 shows the PEG rejection characteristics of the

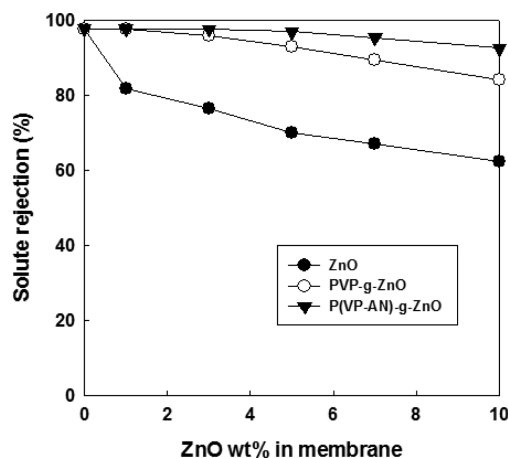


Figure 5. Changes in the PEG rejection of the PES/pristine ZnO, PES/PVP-g-ZnO, and PES/P(VP-AN)-g-ZnO membranes with the ZnO content in the membrane.

PES membranes containing various amounts of pristine ZnO and polymer-grafted ZnO nanoparticles. Note that errors in the water flux did not exceed $\pm 3\%$. The PEG rejection from the PES/pristine ZnO membrane rapidly decreased with increasing ZnO content. However, the PEG rejection of the PES/P(VP-AN)-g-ZnO was nearly identical to that of the PES up to a P(VP-AN)-g-ZnO content of 5 wt %, beyond which there was a slight decrease in the solute rejection with increasing P(VP-AN)-g-ZnO content. The PEG rejection of the PES/PVP-g-ZnO decreased gradually as the PVP-g-ZnO content in the membrane increased. The PEG rejection of the membranes increased in the order PES/pristine ZnO < PES/PVP-ZnO < PES/P(VP-AN)-g-ZnO when the ZnO content was fixed.

The changes in the membrane morphologies and hydrophilicity induced by incorporation of the ZnO nanoparticles were examined to facilitate interpretation of the observed membrane performance. Figure 6 shows FE-SEM images of the cross-sectional morphologies of the membranes. A finger-like structure was observed in the PES membrane when NMP and water was used as the solvent and nonsolvent, respectively.^{5,28}

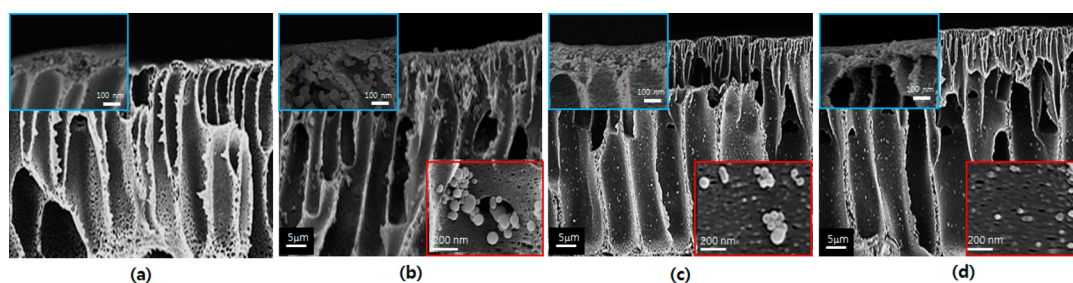


Figure 6. Cross-sectional morphologies of (a) PES, (b) PES/pristine ZnO-5, (c) PES/PVP-g-ZnO-5, and (d) PES/P(VP-AN)-g-ZnO-5 membranes.

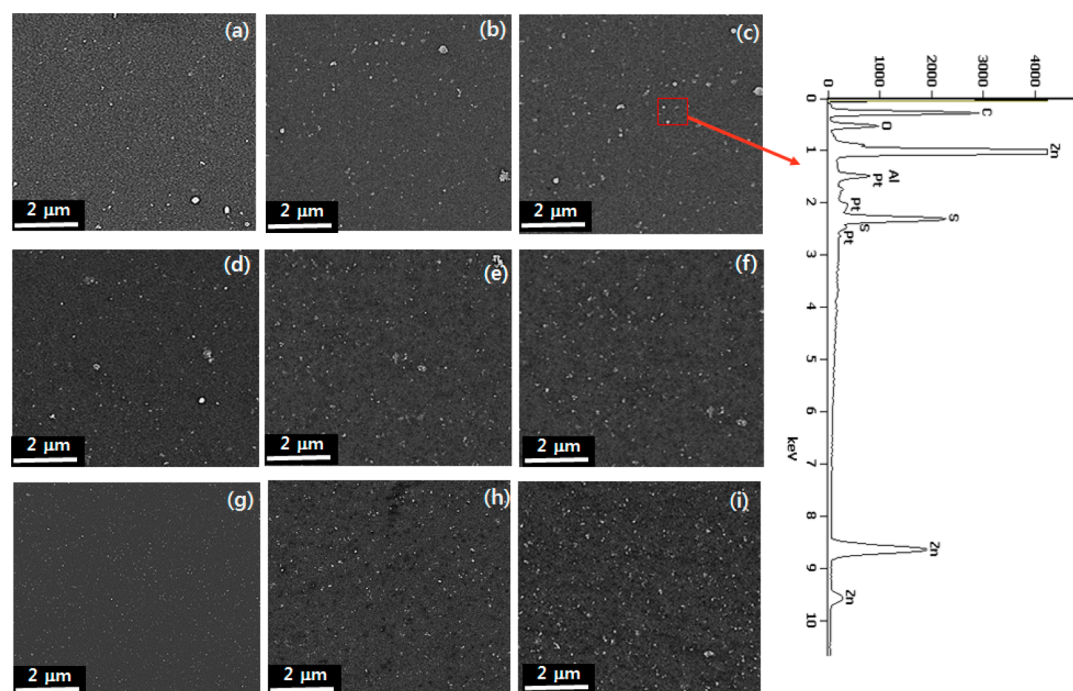


Figure 7. Surface morphologies of (a) PES/pristine ZnO-3, (b) PES/pristine ZnO-5, (c) PES/pristine ZnO-7, (d) PES/PVP-g-ZnO-3, (e) PES/PVP-g-ZnO-5, (f) PES/PVP-g-ZnO-7, (g) PES/P(VP-AN)-g-ZnO-3, (h) PES/P(VP-AN)-g-ZnO-5, and (i) PES/P(VP-AN)-g-ZnO-7 membranes.

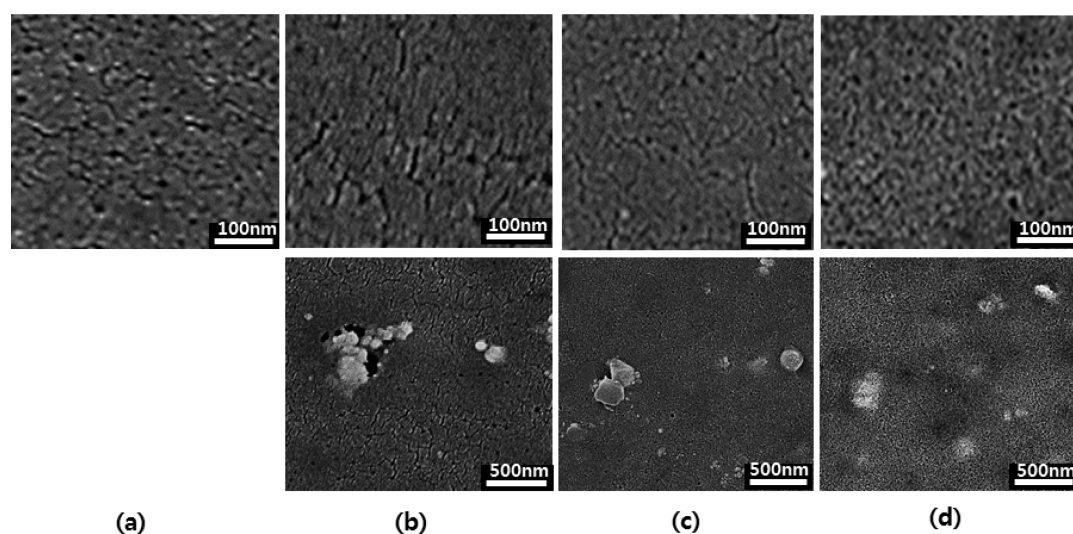


Figure 8. Surface morphologies of (a) PES, (b) PES/pristine ZnO-5, (c) PES/PVP-g-ZnO-5, and (d) PES/P(VP-AN)-g-ZnO-5 membranes.

The PES composite membranes with ZnO nanoparticles also exhibited a finger-like structure. As shown in Figure 6 (blue

boxes), the thickness of the skin-layer was not changed by incorporating ZnO nanoparticles to PES.

Figure 7 shows the surface morphologies of the PES membranes containing pristine ZnO and polymer-grafted ZnO explored with FE-SEM and EDS. Peaks associated with the incorporated ZnO and carbon, sulfur, and oxygen peaks originated from PES were observed on the PES/ZnO composite membrane. The number of ZnO particles existing on the surface increased with increasing ZnO content in the membrane. As shown in Figures 7 and 8, the dispersion of the polymer-grafted ZnO nanoparticles in the PES matrix was better than that of pristine ZnO nanoparticles in the PES matrix. The amount of aggregates on the membrane surface and in the cross-section was reduced in the following order: pristine ZnO > PVP-g-ZnO > P(VP-AN)-g-ZnO. Attractive interactions among the high surface area pristine ZnO nanoparticles led to significant agglomeration of the pristine ZnO nanoparticles in the PES matrix. Aggregation was mitigated by using the polymer-grafted ZnO nanoparticles. When the miscibility between the polymer grafted to the ZnO nanoparticles and PES is higher, the ZnO nanoparticles exhibit better interfacial adhesion with PES along with improved dispersion in the PES matrix. PES forms miscible blends with P(VP-AN) containing 10 wt % AN, but does not form miscible blends with PVP.²⁹ As shown in Figure 8 (high magnification images in the second row), the ZnO dispersion in the PES matrix and the adhesion at the interface between the PES and ZnO nanoparticles improved in the order: pristine ZnO < PVP-g-ZnO < P(VP-AN)-g-ZnO. Overall, the PES membranes containing P(VP-AN)-g-ZnO showed the best level of ZnO dispersion and adhesion at the interface between PES and the ZnO nanoparticles.

When the PES membrane was prepared by the NIPS process, spherical pores (average diameter: 12 nm) and slit-like pores (average width: 4 nm) were formed on the PES membrane surface, as shown in Figure 8a. Spherical pores and slit-like pores were also observed on the PES/ZnO membrane surface where ZnO nanoparticles do not exist, and their size was similar to those of PES membrane (first row in Figure 8b,c,d). Defects (second row in Figure 8b) were observed at the interface between PES and the pristine ZnO aggregates. When defects were formed around the ZnO aggregates in the PES/ZnO membranes, the PEG rejection of the PES/ZnO membrane was reduced. The rapid decline in the PEG rejection with increasing ZnO content in the PES/pristine ZnO membrane arises from defects existing at the interface between the PES matrix and the particle aggregates. The PES/PVP-g-ZnO membrane still contained ZnO aggregates, even though fewer aggregates were formed on the membrane surface (second row in Figure 8c). When compared to the PEG rejection of the PES membrane, a slight reduction in the PEG rejection for the PES/PVP-g-ZnO membrane originated from the PVP-g-ZnO aggregates present on the membrane surface. The PES/P(VP-AN)-g-ZnO membrane exhibited improved dispersion of ZnO nanoparticles in the PES matrix and adhesion at the interface between ZnO and PES when compared to the PES/PVP-g-ZnO membrane (second row in Figure 8d). Thus, the PES/P(VP-AN)-g-ZnO membrane exhibited the highest PEG rejection among the PES/ZnO composite membranes at a fixed ZnO content.

The improved hydrophilicity of the membrane led to an increase in the water flux. The water flux of the PES composite membrane increased with increasing ZnO content. As shown in Figure 9, the contact angle between water droplet and the membrane surface decreased with increasing ZnO content in

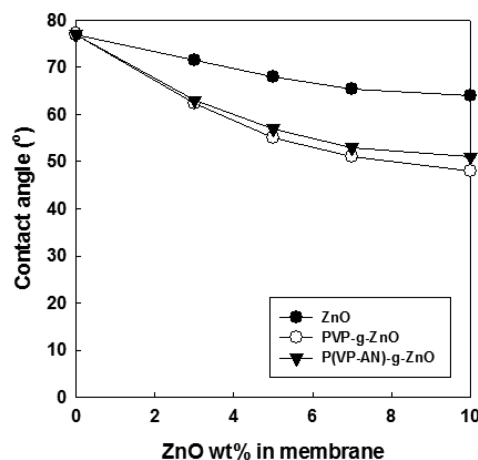


Figure 9. Changes in the water contact angles of the PES/pristine ZnO, PES/PVP-g-ZnO, and PES/P(VP-AN)-g-ZnO membranes as a function of ZnO content.

the membrane. Defects formed at the interface between PES and the pristine ZnO aggregates also induced an increase in the water flux of the membrane. An increase in the water flux of the PES/pristine ZnO membranes with increasing ZnO content arises from the hydrophilic nature provided by pristine ZnO nanoparticles and the defects around the ZnO aggregates. At a fixed ZnO content, the PES/P(VP-AN)-g-ZnO membrane exhibited higher hydrophilicity than the PES/pristine ZnO membrane because of the hydrophilic copolymer grafted to the ZnO nanoparticles. Note that no defects were found at the interface between the P(VP-AN)-g-ZnO and PES matrix. Thus, the enhanced hydrophilicity of the PES/P(VP-AN)-g-ZnO membranes led to higher water flux compared to the PES/pristine ZnO membrane.

Leaching of the ZnO nanoparticles from the PES composite membranes containing 5 wt % of ZnO nanoparticles was examined. The membranes were sonicated in distilled water for 2 h and then the ZnO content in the resulting water was estimated via ICP. Leached ZnO at a concentration of 1.2×10^{-5} g/L was detected for PES/pristine ZnO = 95/5, whereas no leaching was detected for the PES/polymer-grafted ZnO = 95/5 membranes. This result confirmed that the polymer grafted to the ZnO nanoparticles prevents the leaching of ZnO from the membranes.

3.3. Antibacterial Effectiveness and Antifouling Property of Membranes. The antibacterial activity of the membrane was evaluated in accordance with the JIS Z-2801 standard. Antibacterial products require a value of 2.0 or higher antibacterial activity for antibacterial effectiveness.⁴³ The antibacterial test results were presented in Table 2. The antibacterial activity of the PES membrane for both *E. coli* and *S. aureus* was quantified as 0.2. This result indicates that the PES membrane does not have antibacterial effectiveness against both bacteria. For the PES membranes containing 0.1 and 0.3 wt % of pristine ZnO nanoparticles, the antibacterial activities against *E. coli* were 0.7 and 2.7, respectively, while those against *S. aureus* were 2.3 and 4.2, respectively. This result indicates that ZnO nanoparticles incorporated in the PES membrane led to higher antibacterial activity against *S. aureus* than toward *E. coli*. The antibacterial activities against both bacteria increased to 6.1 when the PES membranes contained more than or equal to 0.5 wt % of pristine ZnO. The antibacterial activities of the PES membranes containing PVP-g-ZnO or P(VP-AN)-g-ZnO

Table 2. Antibacterial Activities of PES and PES/ZnO Membranes for both *E. coli* and *S. aureus*

Membrane	Antibacterial Activity (E. coli)	Antibacterial Activity (S. aureus)	Image for E. Coli	Image for S. aureus
PES	0.2	0.2		
PES/ZnO = 99.9/0.1	0.7	2.3		
PES/ZnO = 99.7/0.3	2.7	4.2		
PES/ZnO = 99.5/0.5	6.1	6.1		
PES/ZnO = 99/1	6.1	6.1		
PES/PVP-g-ZnO = 99.5/0.5	6.1	6.1		
PES/P(VP-AN)-g-ZnO = 99.5/0.5	6.1	6.1		

nanoparticles were identical to those of the PES membranes containing pristine ZnO nanoparticles, as shown in Table 2. This result indicates that the polymers grafted to the ZnO nanoparticles do not influence the antibacterial activity of the membrane. It is known that the antibacterial activity of ZnO is derived from the hydrogen peroxide generated at the ZnO surface.³⁶ Consequently, an inhibitory effect on the growth of bacteria can be provided to the PES membrane by incorporating ZnO nanoparticles, and the utilization of more ZnO should lead to better antibacterial effectiveness.

Figure 10 shows changes in the permeate flux of the membranes with operating time. Note that an aqueous solution containing 100 times the CMC of $C_{16}E_8$ was used as the feed solution. A rapid decrease in the permeate flux of the PES membrane was observed during the first 20 min and then leveled off owing to adsorbed surfactant on the membrane surface and in the pores. The permeate flux of the PES membranes containing ZnO also reduced during about the first 130 min and then leveled off. After 210 min, the reduction percentage in the permeate flux for the PES, PES/ZnO = 95/5, PES/PVP-g-ZnO = 95/5, and PES/P(VP-AN)-g-ZnO = 95/5 membranes reached 80.0%, 54.5%, 39.5%, and 39.2%, respectively. The permeate flux of the PES membranes containing ZnO nanoparticles declined to a lesser extent with time when compared to the PES membrane. The increased hydrophilicity derived from incorporation of the ZnO nanoparticles enhanced the antifouling properties of the membranes. The decrease in the permeate flux of the PES membranes containing polymer-grafted ZnO with time was smaller than that of the PES membranes containing pristine ZnO. These results suggest that fouling of the membranes was further reduced by the hydrophilic polymers grafted onto the ZnO

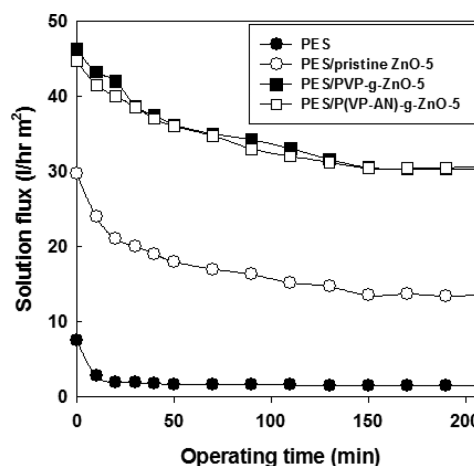


Figure 10. Changes in the solution flux of the PES, PES/pristine ZnO-5, PES/PVP-g-ZnO-5, and PES/P(VP-AN)-g-ZnO-5 membranes with respect to the operating time.

nanoparticles. As a summary, ultrafiltration membranes exhibiting antibacterial effectiveness, improved water flux, and antifouling characteristic without changes in solute rejection could be produced by incorporating P(VP-AN) copolymer-grafted ZnO nanoparticles into the PES membrane matrix.

4. SUMMARY

Ultrafiltration membranes exhibiting antibacterial effectiveness, an improved water flux, and excellent antifouling characteristic were fabricated from PES and ZnO grafted with P(VP-AN) or PVP by the NIPS process. The ZnO dispersion in the PES

matrix and the adhesion at the interface between PES and ZnO increased in the following order: pristine ZnO < PVP-g-ZnO < P(VP-AN)-g-ZnO. Although the unmodified PES membrane did not exhibit antibacterial effectiveness, the PES membranes containing more than 0.5 wt % of ZnO nanoparticles exhibited antibacterial activity (= 6.1) toward *E. coli* and *S. aureus*, regardless of the polymer grafted onto the ZnO. The hydrophilicity of the membranes increased with increasing ZnO content, and the PES membrane containing polymer-grafted ZnO showed better hydrophilic nature than the PES membrane with pristine ZnO. The water flux of the PES/P(VP-AN)-g-ZnO membrane increased with increasing filler content without changes in the solute rejection when the P(VP-AN)-g-ZnO concentration was 5 wt % or less. The PES membrane containing ZnO nanoparticles also showed improved antifouling characteristic when compared to the PES membrane only, whereas the PES membrane containing polymer-grafted ZnO nanoparticles showed better antifouling characteristic than the PES/pristine ZnO membrane. Overall, membranes with antibacterial effectiveness, improved water flux, and antifouling characteristic could be produced from PES and P(VP-AN) copolymer-grafted ZnO nanoparticles.

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This research was supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF), funded by the Ministry of Education, Science, and Technology (2015R1A2A2A01003230). Additional support was provided by the Chung-Ang University Excellent Student Scholarship (2015).

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